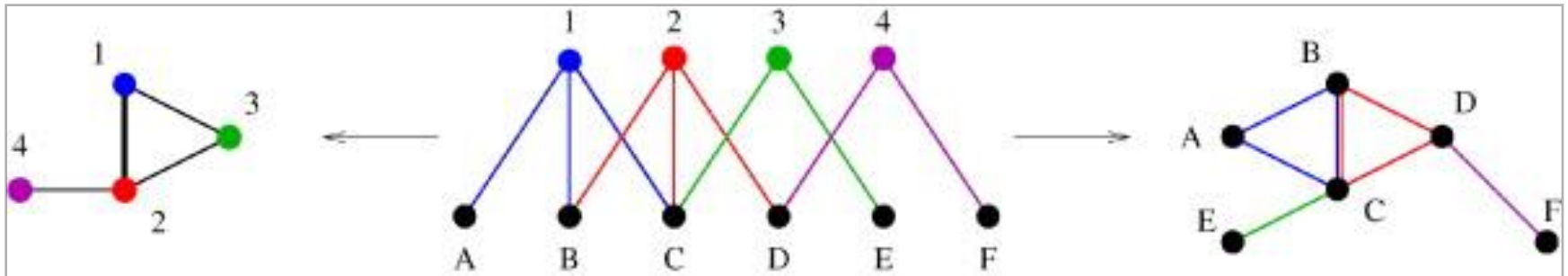


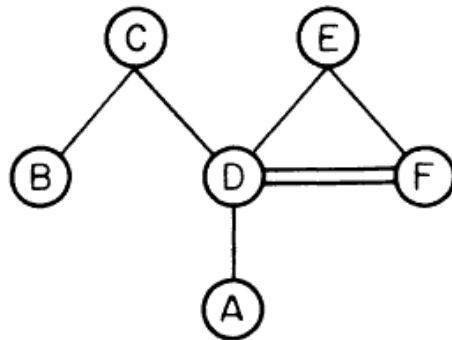
A two-mode network and its one-mode projections

- A two-mode (or affiliation) network consists of:
 - Two types of nodes.
 - There are edges that make connections **between** types but not **within** types.
- Examples
 - Actors and movies.
 - Scientists and papers.
 - Persons and Events.

A two-mode network and its one-mode projections



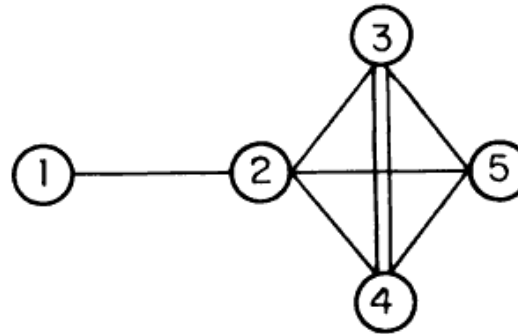
A two-mode network and its one-mode projections



A-1. Interpersonal network

	A	B	C	D	E	F
A	0	0	0	1	0	0
B	0	0	1	0	0	0
C	0	1	0	1	0	0
D	1	0	1	0	1	2
E	0	0	0	1	0	1
F	0	0	0	2	1	0

B-1. Matrix representation (P) of Figure 1. A-1



A-2. Intergroup network

	1	2	3	4	5
1	0	1	0	0	0
2	1	0	1	1	1
3	0	1	0	2	1
4	0	1	2	0	1
5	0	1	1	1	0

B-2. Matrix representation (g) of Figure 1. A-2

	1	2	3	4	5
A	0	0	0	0	1
B	1	0	0	0	0
C	1	1	0	0	0
D	0	1	1	1	1
E	0	0	1	0	0
F	0	0	1	1	0

C. The binary adjacency matrix (A) of person-to-group affiliations

Figure 1. Fictitious Data

A two-mode network and its one-mode projections (Breiger 1974: 182).

Although these differences between the interpersonal network and the intergroup network are quite evident, the P and G matrices nonetheless stand in intimate relation to one another. Following Simmel (1955:125–8, 147), think of each tie between two groups as a set of persons who form the “intersection” of the groups’ memberships. In the dual case, think of each membership tie between two persons as the set of groups in the “intersection” of *their* individual affiliations.

A two-mode network and its one-mode projections (Breiger 1974: 182).

$$P_{ij} = \sum_{k=1}^g A_{ik}A_{jk} \quad (1)$$

and similarly for ties between groups:

$$G_{ij} = \sum_{k=1}^p A_{ki}A_{kj} \quad (2)$$

The matrix A^T of group-to-person ties is equivalent to A except that its rows are interchanged or “transposed” with its columns; that is, A^T is of dimension $g \times p$ and $A^T_{ij} = A_{ji}$ for any i and j . Hence we may rewrite the above equations using the person-to-group “translation” matrix A and its transpose to obtain the fundamental equalities:

$$P = A(A^T) \quad (3)$$

$$G = (A^T)A \quad (4)$$

where the multiplication is ordinary (inner product) matrix multiplication. Thus: two distinct matrices, one of person-to-person relations (P) and one of group-to-group relations (G), are uniquely defined by and derivable from a single “translation” matrix (A) of person-to-group affiliations.³

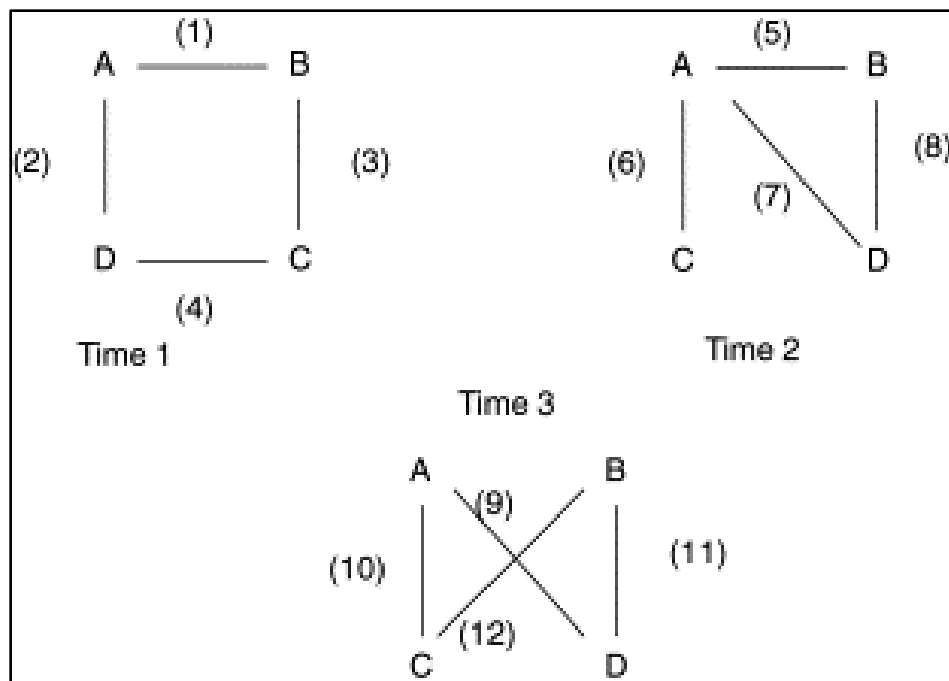
Advantages of the two-mode to one-mode projection

- We can analyze the network using all of the tricks that we have learned so far :
 - We can study characteristics of the “**ego-network**” of one of the node types (e.g. the diversity of organizations that a person belongs to or the diversity of membership in an organization [McPherson 1982; McPherson and Smith-Lovin 1987]).
 - We can calculate **centrality measures** for each node type (Bonacich 1991)).
 - We can find **cohesive subgroups**.

Disadvantages of the two-mode to one-mode projection

- Information encoded in the bipartite structure is **lost** after projection.
- **Edge inflation**: The number of links in the one-mode graph is much larger than in the bipartite graph.
- **Artifactual** properties: clustering the one-mode projection may not reflect actual clustering in the two-mode network.

Hyperedges: Another way to represent data in “network” form



→ $E =$

A	B	C	D	t1	t2	t3
1	1	0	0	1	0	0
1	0	0	1	1	0	0
0	1	1	0	1	0	0
0	0	1	1	1	0	0
1	1	0	0	0	1	0
1	0	1	0	0	1	0
1	0	0	1	0	1	0
0	1	0	1	0	1	0
1	0	0	1	0	0	1
1	0	1	0	0	0	1
0	1	0	1	0	0	1
0	1	1	0	0	0	1

A two-mode network and its one-mode projections

	Actor-Movies	Authoring	Occurrences	Peer-to-peer
Number of links in G	1,470,418	45,904	183,363	55,829,392
Number of links in G(bottom)	15,038,083	29,552	392,066	10,142,780,673
Number of links in G(top)	20,490,112	134,492	51,405,275	1,085,217,140

Latapy et al's Basic Statistics in Two-Mode Networks

- Number of (top) Nodes: $n_{\uparrow}$
- Number of (bottom) Nodes: $n_{\downarrow}$
- Number of edges: m .
- Average (top) degree: $k_{\uparrow} = m/n_{\uparrow}$
- Average (bottom) degree: $k_{\downarrow} = m/n_{\downarrow}$
- Average Degree in the “Bipartite” graph:
 - $k = (2m/(n_{\uparrow} + n_{\downarrow}))$
- Density in the two-mode graph:
 - $\delta(G) = (m / (n_{\uparrow} n_{\downarrow}))$
- Density in the “Bipartite” Graph:
 - $\delta(G') = 2m / ((n_{\uparrow} + n_{\downarrow})(n_{\uparrow} + n_{\downarrow} - 1))$

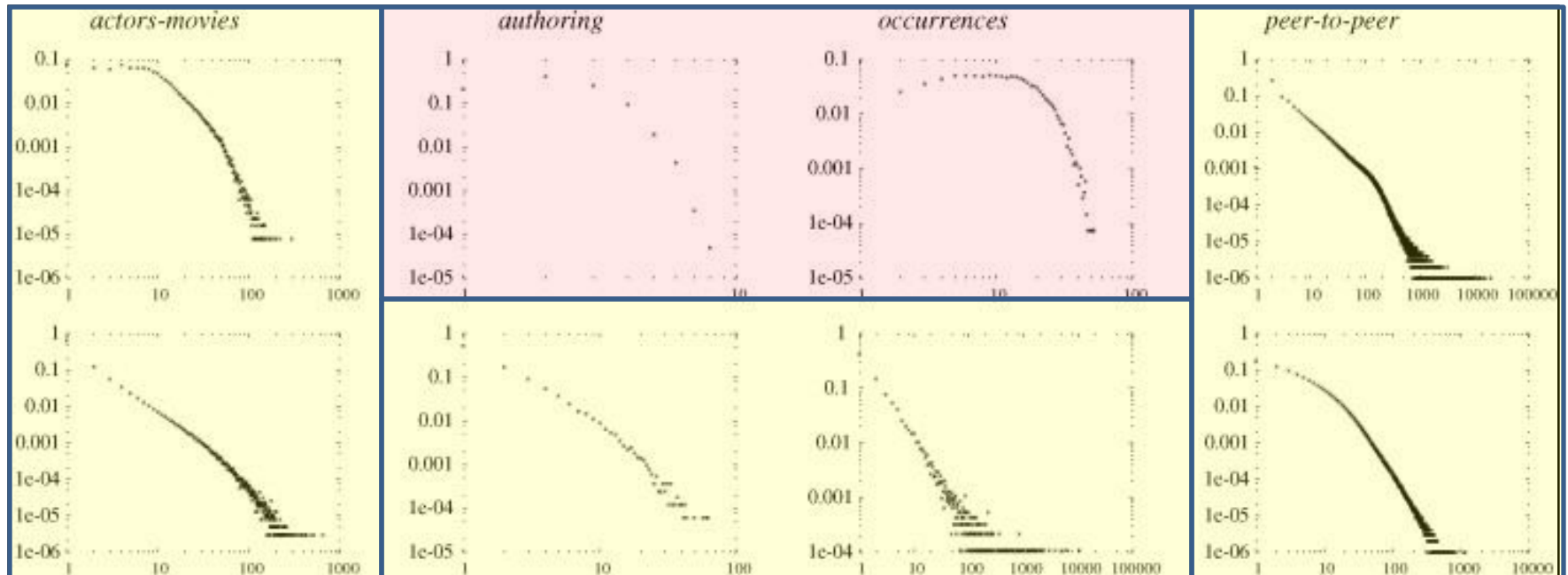
The bipartiate graph

$$\begin{array}{c} \text{Groups} \\ \text{Persons} \end{array} \begin{array}{cc} \text{Groups} & \text{Persons} \\ \left(\begin{array}{cc} 0 & A^t \\ A & 0 \end{array} \right) \end{array}$$

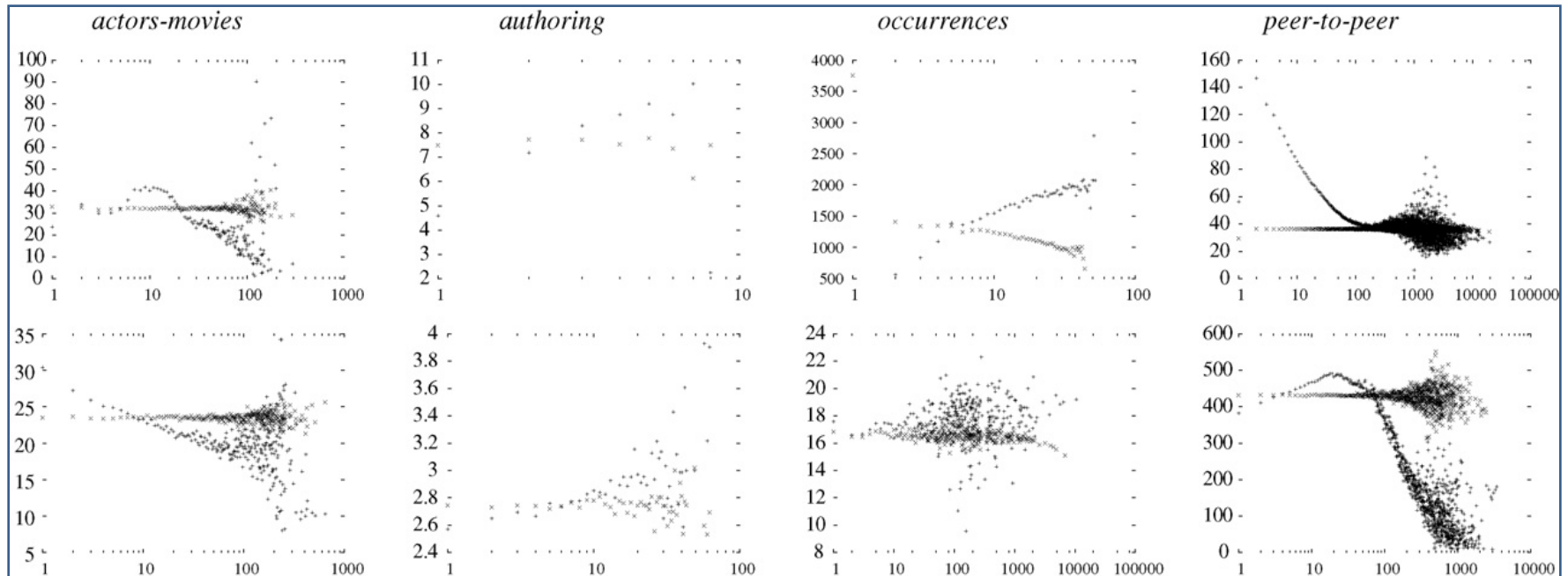
Basic Statistics in two-mode networks

	Actor-Movies		Authoring		Occurrences		Peer-to-peer	
	Real	Random	Real	Random	Real	Random	Real	Random
n(top)	127,823	idem	19,885	idem	13,587	idem	1,986,588	idem
n(bottom)	383,640	idem	16,400	idem	9,264	idem	5,380,546	idem
m	1,470,418	idem	45,904	idem	183,363	idem	55,829,392	idem
k(top)	11.5	idem	2.3	idem	13.5	idem	28.1	idem
k(bottom)	3.8	idem	2.8	idem	19.8	idem	10.4	idem
k	5.7	idem	2.5	idem	16	idem	15.2	idem
δ	0.00003	idem	0.00014	idem	0.0015	idem	0.0000052	idem
lcc(top)	124,414	125,944	16,209	18,512	13,579	13,587	1,986,343	1,426,978
lcc(bottom)	374,511	381,431	11,654	14,607	9,246	9,264	5,380,507	5,054,689
d(top)	6.8	5.3	13.1	9.3	3.1	3	5.3	5
d(bottom)	7.3	5.8	13.9	9.9	3.8	3.7	5.4	4.9
d	7.2	5.8	13.5	9.6	3.4	3.2	5.3	4.9

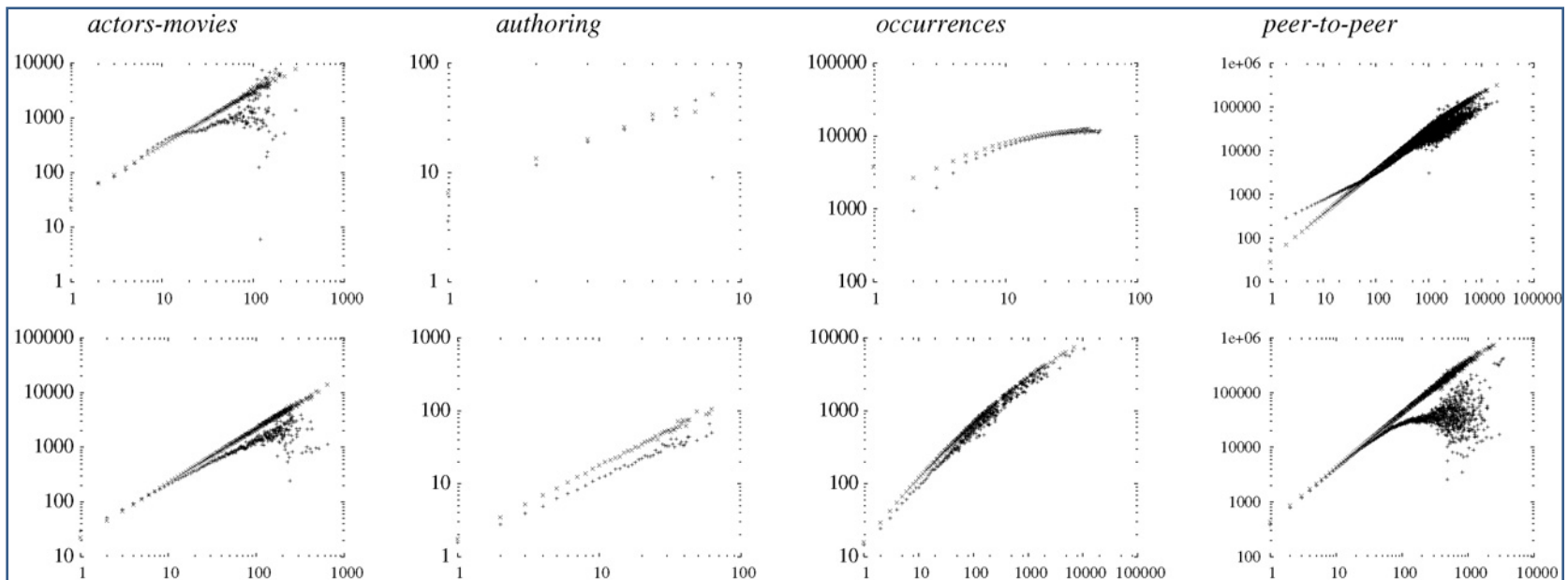
Degree Distributions in two-mode networks



Degree-Correlations in Two-Mode Networks

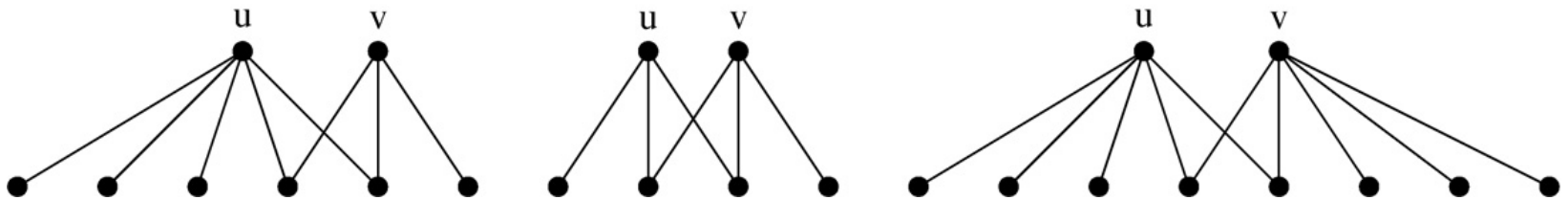


Degree against distance-2 Neighbors



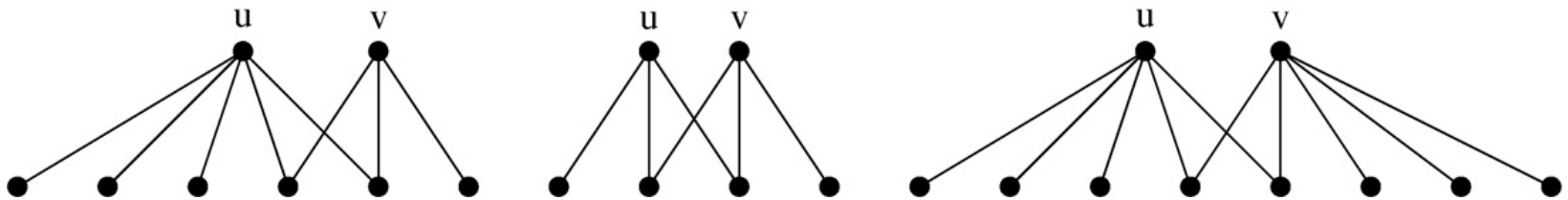
Bipartite Clustering

- First definition: The clustering between any two nodes which belong to the same set (top or bottom) is given by the intersection of their one-step neighborhood divided by their union.



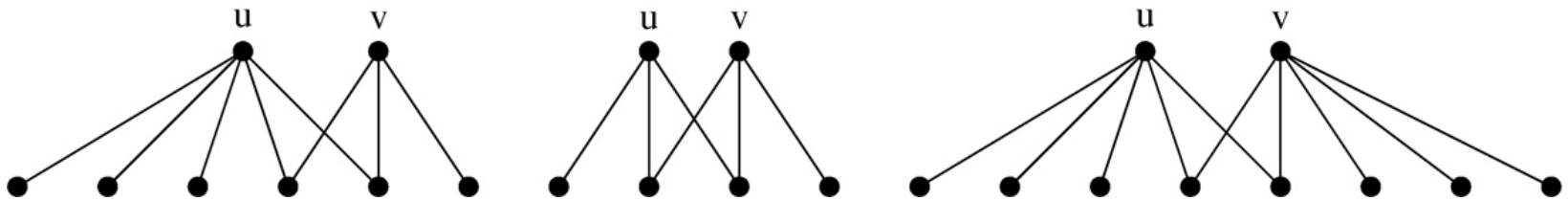
Bipartite Clustering

- First definition: The clustering coefficient between any two nodes $[cc.(u, v)]$ which belong to the same set (top or bottom) is given by the intersection of their one-step neighborhood divided by their union.



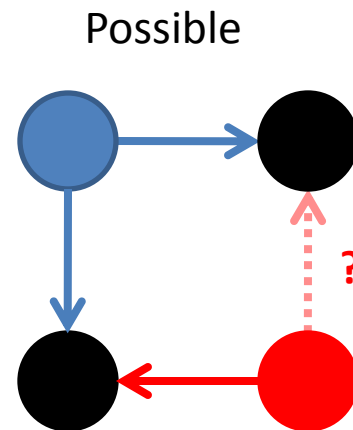
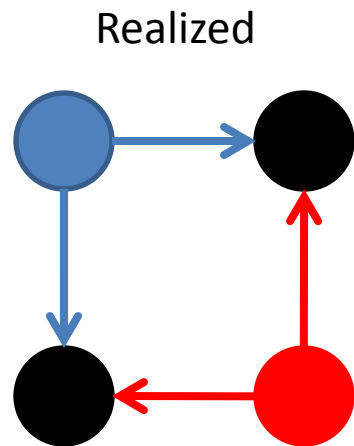
Bipartite Clustering

- A generalization: for any (top or bottom) node u , its clustering coefficient ($cc.(↑)$ or $cc.(↓)$) is given by its **average overlap** with other nodes of the same type (v_j) for which the overlap does not equal zero.
- Total graph clustering:
 - $cc(G) = (n↑cc.(↑) + n↓cc.(↓)) / (n↑n↓)$



Bipartite Clustering

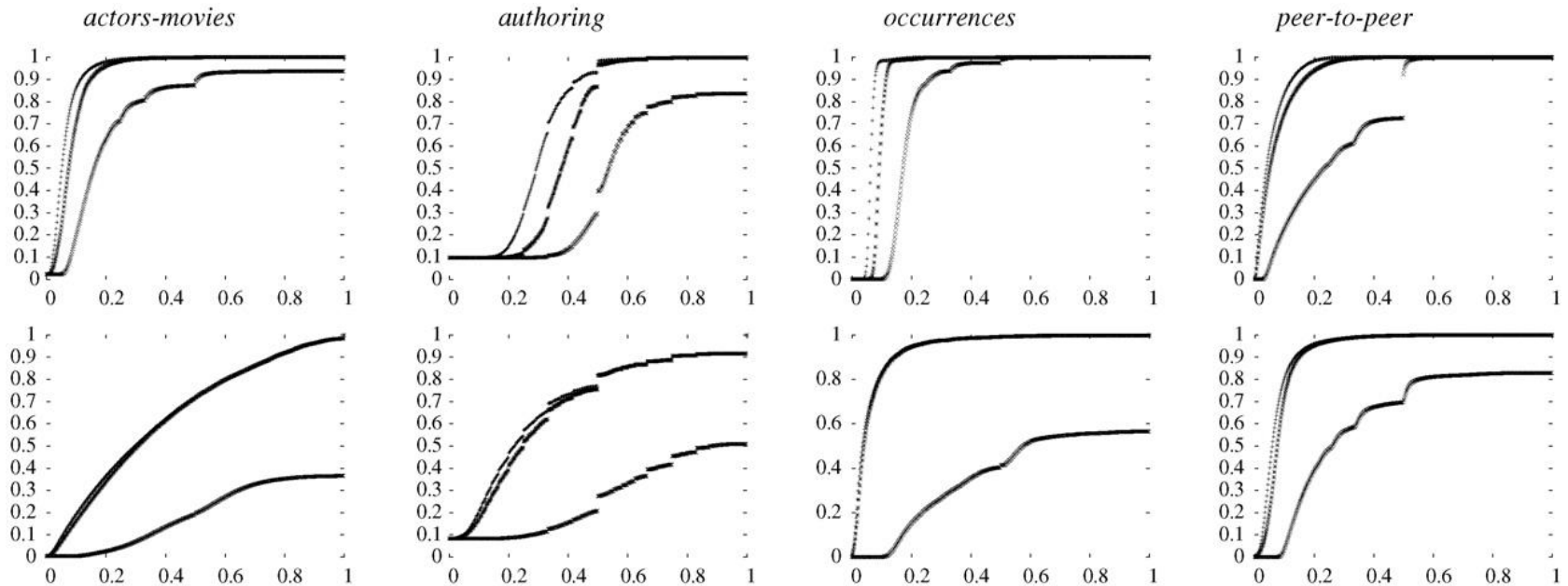
- A probabilistic formulation of two-mode clustering:
 - $CCN(G) = (\text{number of realized quadruplets}) / (\text{number of possible ones})$



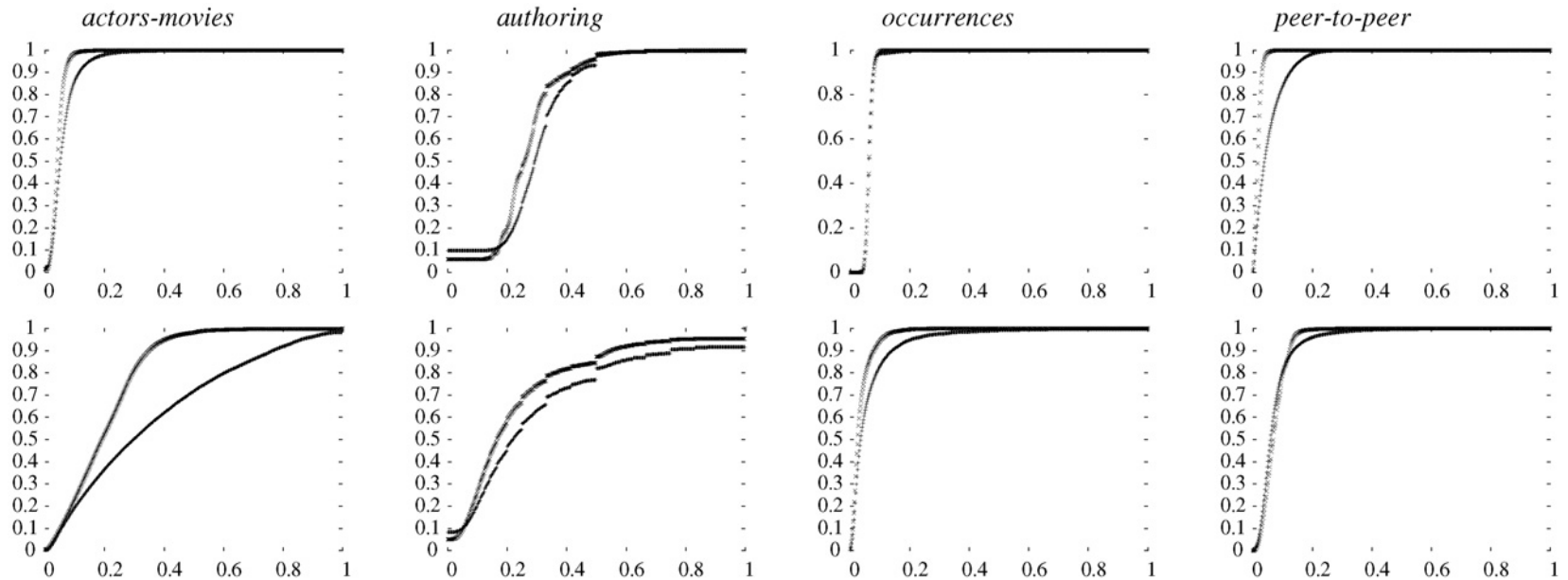
Clustering in two-mode networks

	Actor-Movies		Authoring		Occurrences		Peer-to-peer	
	Real	Random	Real	Random	Real	Random	Real	Random
cc•(top)	0.064	0.046	0.29	0.27	0.066	0.066	0.056	0.019
cc•(bottom)	0.36	0.2	0.31	0.25	0.065	0.038	0.076	0.074
CC(N)	0.0082	0.00024	0.079	0.00012	0.053	0.048	0.0094	0.00019
cc•_min(top)	0.24	0.23	0.56	0.56	0.19	0.2	0.27	0.24
cc•_min(bottom)	0.81	0.79	0.73	0.7	0.64	0.61	0.39	0.42
cc•_max(top)	0.087	0.062	0.36	0.34	0.097	0.097	0.074	0.024
cc•_max(bottom)	0.37	0.21	0.33	0.26	0.069	0.041	0.091	0.089

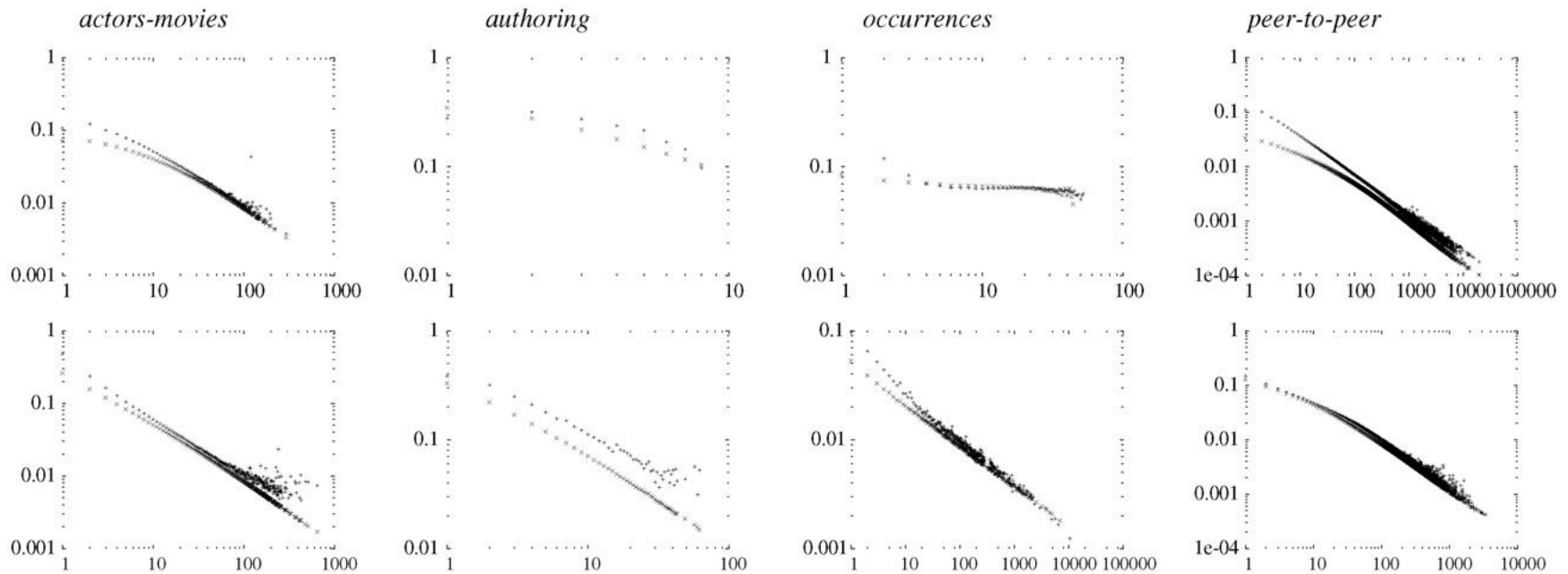
Cumulative Distribution of Clustering in two-mode networks



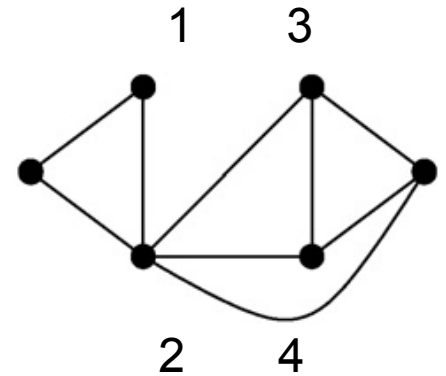
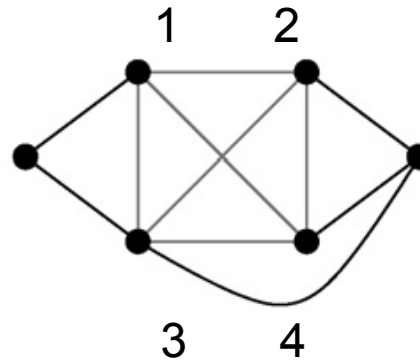
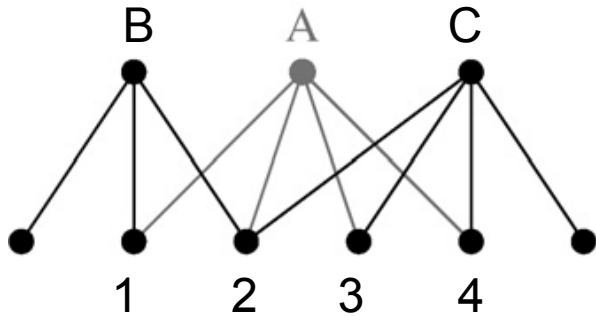
Cumulative Distribution of Clustering in two-mode networks (Versus Random)



Correlation between node degree and clustering



Redundancy in two mode networks



	<i>B</i>	<i>C</i>
<i>1</i>	1	0
<i>2</i>	1	1
<i>3</i>	0	1
<i>4</i>	0	1



	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>
<i>1</i>	0	1	0	0
<i>2</i>	1	0	1	1
<i>3</i>	0	1	0	1
<i>4</i>	0	1	1	0

$$4/6 = 0.66$$

Redundancy Cumulative Distribution (versus random)

